GRAPH
PROBLEM 6

QUESTION:

DS	**Session**	Graph	**Question Information**	Level 1 Challenge 86

Question description

There are n boys and m girls in a school. Next week a school dance will be organized. A dancing pair consists of a boy and a girl, and there are k potential pairs.

Your task is to find out the maximum number of dance pairs and show how this number can be achieved.

Constraints

- $1 \le n,m \le 500$
- $1 \le k \le 1000$
- $1 \le a \le n$
- $1 \le b \le m$

Input

The first input line has three integers n, m and k: the number of boys, girls, and potential pairs. The boys are numbered 1,2,…,n, and the girls are numbered 1,2,…,m.

After this, there are k lines describing the potential pairs. Each line has two integers a and b: boy a and girl b are willing to dance together.

Output

First print one integer *rr*: the maximum number of dance pairs. After this, print lines describing the pairs. You can print any valid solution.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
3 2 4
1 1
1 2
2 1
3 1
```

EXPECTED OUTPUT

```
2
1 2
2 1
```

Test Case 2

INPUT (STDIN)

```
3 2 4
1 1
1 2
2 3
3 2
```

EXPECTED OUTPUT

```
3
1 1
2 3
3 2
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
void link(int u,int v)
```

Test Case 2

KEYWORD

```
int bfs(int n)
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

```
2
```

Test Case 2

TOKEN COUNT

```
600
```

Test Case 3

NLOC

```
84
```

SOURCE CODE:

```c
#include <stdio.h>
#include <stdbool.h>

#define MAX_PEOPLE 505

int partner_boy[MAX_PEOPLE];
int partner_girl[MAX_PEOPLE];
bool girl_inspected[MAX_PEOPLE];
int willingness[MAX_PEOPLE][MAX_PEOPLE];
int willing_count[MAX_PEOPLE];

bool augment_pair(int boy) {
    for (int idx = 0; idx < willing_count[boy]; idx++) {
        int girl = willingness[boy][idx];
        if (!girl_inspected[girl]) {
            girl_inspected[girl] = true;
            if (partner_girl[girl] == 0 || augment_pair(partner_girl[girl])) {
                partner_girl[girl] = boy;
                partner_boy[boy] = girl;
                return true;
            }
        }
    }
    return false;
}

int main() {
    int total_boys, total_girls, offers;
    if (scanf("%d %d %d", &total_boys, &total_girls, &offers) != 3) return 0;

    for (int i = 0; i < offers; i++) {
        int b, g;
        scanf("%d %d", &b, &g);
        willingness[b][willing_count[b]++] = g;
    }

    int matched_couples = 0;
    for (int b = 1; b <= total_boys; b++) {
        for (int g = 1; g <= total_girls; g++) girl_inspected[g] = false;
        if (augment_pair(b)) matched_couples++;
    }

    printf("%d\n", matched_couples);
    for (int b = 1; b <= total_boys; b++) {
        if (partner_boy[b] > 0) {
            printf("%d %d\n", b, partner_boy[b]);
        }
    }
    return 0;
}
```

OUTPUT:

```
3 2 4
1 1
1 2
2 1
3 1
2
1 2
3 1
```

```
3 2 4
1 1
1 2
2 3
3 2
3
1 1
2 3
3 2
```